



Hohete Tibebe Share Company
Education and Technology Team
Ethio-Parents' Schools: Addis Ababa (Gerji/Gullele) and Hawassa.
2nd Semester Mathematics Final Examination For Grade 9



Time Allowed 1:30 hour

June 2020

Max. Marks 40%

PART I: TRUE OR FALSE ITEMS (1 PTEACH)

WRITE TRUE IF THE STATEMENT IS CORRECT AND WRITE FALSE IF THE STATEMENT IS WRONG

1. For a regular polygon with n-number of sides, then it has n, number of line of symmetries.
2. The angle at each vertex of a polygon, the interior & exterior angles are supplementary.
3. The quadratic function $y = 4x^2 + 2x + 1$ has a maximum value.
4. Two triangles are congruent if they have the same size.
5. The degree measure of $\frac{\pi}{4}$ radian is 45° .
6. A sector of a given circle is formed by a chord and radii of the circle
7. Variance of a set of data can never be negative under any condition.
8. If A is an impossible event, then $0 < p(A) < 1$.
9. For any event E , $0 < p(E) < 1$.
10. The radian measure of 0° is 0 radian.

PART II: MULTIPLE CHOICE ITEMS (1 PTEACH)

CHOOSE THE CORRECT ANSWER FOR THE FOLLOWING QUESTIONS.

11. What is the area of a regular hexagon inscribed in a circle of radius 12cm?
A. $216\sqrt{3}$ B. 216 C. 72 D. $72\sqrt{3}$
12. In a regular polygon, the degree measure of each central angle is 30° . What is the sum of all interior angles of such a polygon?
A. 1800 B. 2161 C. 1440 D. 1980
13. Two dies are tossed simultaneously. What will be the probability that sum of two numbers on the upper faces is less than 4?
A. $\frac{2}{3}$ B. $\frac{1}{12}$ C. $\frac{1}{6}$ D. $\frac{1}{9}$

14. Let $A = \{1, 2, 3, 4\}$ and $B = \{-1, 0, 1, 2\}$ be given two sets. If $R = \{(x, y) : y = x - 1, x \in A \text{ \& } y \in B\}$, then which one of the following set contains order pair of R ?
- A. $\{(1, 0), (2, 1), (3, 2)\}$ C. $\{(0, 1), (1, 2), (2, 3)\}$
 B. $\{(-1, -2), (0, -1), (1, 0), (2, 1)\}$ D. $\{(1, 0), (2, 1), (3, 2), (4, 3)\}$
15. Let $A = \{x : 1 \leq x < 8\}$, where $x \in \mathbb{Z}$ and $B = \{2, 4, 6, 8\}$. If $R = \{(x, y) : 3x - 2y > 3\}$ is a relation from A to B given then what is the domain of R ,
- A. $\{2, 4, 6, 8\}$ C. $\{1, 2, 3, 4, 5, 6, 7\}$
 B. $\emptyset$ D. $\{3, 4, 5, 6, 7\}$
16. Which of the following is a function?
- A. $R = \{(x, y) / x = y^2\}$ C. $\{(-1, 5), (4, 6), (-1, 9), (0, 0)\}$
 B. $R = \{(1, 1), (2, 1), (5, 1), (6, 2)\}$ D. $R = \{(x, y) : x = 3\}$
17. If $g = \{(2, 1), (5, 3), (1, 2), (0, -1)\}$ and $f = \{(5, 1), (6, -2), (1, 5), (3, 0)\}$, then which one is $f \circ g$ is equal to:
- A. $\{(-2, 5), (3, 1)\}$ C. $\{(0, -2), (0, 3)\}$
 B. $\{(5, -2), (1, 3)\}$ D. $\{(5, 2), (1, -3)\}$
18. Let $f(x) = -3x - 12$ and $g(x) = x - 2$, then $\left(3f + 5\frac{f}{g}\right)(0)$ is equal to
- A. -6 C. -20
 B. -66 D. -30
19. Which of the following is not true about the graph of the function $f(x) = -3x^2 + 9$?
- A. Its x-intercepts are $-\sqrt{3}$ and $\sqrt{3}$ C. Its line of symmetry is $x = 0$.
 B. Its vertex is $(0, 9)$ D. Its range is $(-\infty, 9)$.
20. Let $f(x) = -x^2 + 4x$, then the vertex of the parabola is at
- A. $(1, 4)$ C. $(0, 4)$
 B. $(0, 0)$ D. $(-4, 32)$

21. Let $R = \{ (x, y) : y \geq x \text{ and } y + x < 2 \}$ which one of the following is **not true**?
- A. $(-1, 0) \in R$ C. $(1, 1)$ is not belongs to R
 B. Range of R is $(-\infty, 1)$ D. Domain of R is $(-\infty, 1)$

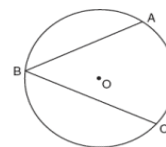
22. The degree measure of each interior angle of a 15 sided regular polygon is
- A. 243° C. 172°
 B. 156° D. 24°

23. The sum of the measures of the three interior angles of scalene triangle is:-
- A. 360° C. 180°
 B. 0° D. 90°

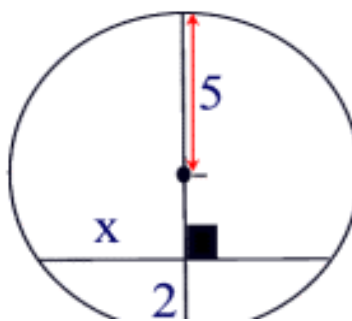
24. The apothem of an inscribed regular hexagon is 6cm. What is the area of the Hexagon?
- A. $144\sqrt{3} \text{ cm}^2$ C. $36\sqrt{3} \text{ cm}^2$
 B. $72\sqrt{3} \text{ cm}^2$ D. $12\sqrt{3} \text{ cm}^2$

25. In the accompanying diagram below of circle O , $m\angle ABC = 2x$ and $m\widehat{AC} = x + 60$.
 Then the value of x is _____

- A. 20° B. 60° C. 40° D. 80°



26. Given the circle at the right with designated center, designated perpendicular, and $r = 5\text{cm}$



Find the length of the segment labeled by x .

- A. 4cm B. 5cm C. 8 cm D. $\sqrt{5} \text{ cm}$

27. If three fair coins are tossed, what is the probability of getting *at least* two heads?
- A. $\frac{2}{3}$ B. $\frac{1}{2}$ C. $\frac{3}{8}$ D. $\frac{1}{8}$

28. Which of the following formulas will find the total surface area (SA) of a cylinder which is open at one end?

A. $SA = 2\pi rh + 2\pi r^2$

C. $SA = 2\pi rh - \pi r^2$

B. $SA = 2\pi rh + \pi r^2$

D. $SA = 2\pi rh$

29. Maximum possible exterior angle in a regular polygon is _____.

A. 70°

B. 60°

C. 90°

D. 120°

30. $\triangle ABC \sim \triangle DEF$, if the area of triangle ABC is 64 sq.cm and area of DEF is 9 sq.cm.

If AB = 24cm. Then DE = _____

A. 6cm

B. 4cm

C. 9cm

D. 10cm

PART III: SHORT ANSWER QUESTIONS ITEMS

WRITE THE SIMPLIFIED ANSWER ON THE ANSWER SHEET PROVIDE

31. The domain of the relation $R = \{(x, y) : y = \sqrt{x^2 - 1}\}$ is _____

32. The ratio of the perimeter of a regular hexagon to its radius is _____.

33. The area of a regular hexagon whose side 2 cm long is _____

34. Sum of all interior angles of a polygon with (n) sides is given by _____.

35. What is a polygon called if the sum of its interior angles equals 1440° ? _____.

PART IV: WORKOUT QUESTIONS ITEMS

SHOW THE NECESSARILY STEPS CLEARLY

36. The following table is the frequency distribution of a certain set of data.

v	10	11	12	13	14	15
F	2	0	x	2	4	2

If the mean of this data is 12.8, then the value of x is _____

37. Find the area of a regular 8 sided polygons with radius 4cm

38. The area of a trapezium with lower base 12 and upper base 16 and the two bases are 6cm apart each other.

39. Show and shade the region on the Venn-diagram for any two sets A and B, that is $(A \cup B)'$ is equal to:-

40. Let $R = \{(x, y) / 2x - 3y > 3\}$ then list few four order pairs which belongs to the relation to R?

===== **GOOD LUCK** =====